Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Python Data Analysis Masterclass

Course Code: C193

Conducted by

Tertiary Infotech Academy Pte Ltd

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial release: two-day lesson plan and learner guide aligned to the 141-slide deck and the ten Python labs.	Dr Alfred Ang

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Course Overview

Python Data Analysis Masterclass is a two-day, 15-hour hands-on course. Learners use Python, pandas, Matplotlib, Seaborn and scikit-learn across ten realistic synthetic-data labs covering preparation, transformation, visualisation, statistics, time series, pipelines and linear regression.

Course Code	C193
Training Duration	15 hours across 2 days (9:30am – 5:30pm)
Trainer	Dr Alfred Ang
Mode	Instructor-led classroom with individual Python labs

Learning Outcomes

- LO1: Prepare data for analysis with pandas.
- LO2: Transform and integrate data for analysis.
- LO3: Gain insights from clear, truthful data visualizations.
- LO4: Perform statistical and time-series analysis.
- LO5: Perform advanced analysis and interpret results credibly.

Daily Schedule

Day 1: Prepare, transform and visualise trustworthy data

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Welcome, introductions and outcomes	Facilitated briefing	—
9:50-11:00	70 min	Topic 1: pandas structures, import, selection, filtering and cleaning	Concept briefing, worked examples and questioning	Slides 10-34
11:00-12:45	105 min	Labs 1-2: prepare retail and customer data	Demonstration, individual practice and debrief	Slides 32-33
12:45-13:15	30 min	Lunch break	Lunch	—
13:15-14:10	55 min	Topic 2: concat, merge, groupby, aggregation and pivot	Concept briefing, worked examples and questioning	Slides 35-59
14:10-15:05	55 min	Labs 3-4: integrate and summarise operations data	Demonstration, individual practice and debrief	Slides 57-58
15:05-15:55	50 min	Topic 3: Matplotlib, Seaborn and visual integrity	Concept briefing, worked examples and questioning	Slides 60-85
15:55-17:10	75 min	Labs 5-6: management visualisation and relationship analysis	Demonstration, individual practice and debrief	Slides 83-84
17:10-17:30	20 min	Day 1 reflection and Q&A	Reflection and Q&A	—

Day total: 7.5 instructional hours (9:30am – 5:30pm), plus a 30-minute lunch.

Day 2: Analyse statistics and time series, then build a credible advanced pipeline

Time	Duration	Topic / Activity	Method	Slides
9:30-9:50	20 min	Day 1 recap	Facilitated briefing	—
9:50-11:25	95 min	Topic 4: descriptive statistics, correlation and time series	Concept briefing, worked examples and questioning	Slides 86-113
11:25-12:45	80 min	Labs 7-8: distributions	Demonstration,	Slides 111-112

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		and energy-demand time series	individual practice and debrief	
12:45-13:15	30 min	Lunch break	Lunch	—
13:15-14:50	95 min	Labs 7-8 (continued): distributions and energy-demand time series	Demonstration, individual practice and debrief	Slides 111-112
14:50-15:50	60 min	Topic 5: pipe, apply, regression and credibility review	Concept briefing, worked examples and questioning	Slides 114-138
15:50-17:15	85 min	Labs 9-10: reusable pipeline and public-health capstone	Demonstration, individual practice and debrief	Slides 136-137
17:15-17:30	15 min	Course recap, Q&A and course feedback	Reflection and Q&A	—

Day total: 7.5 instructional hours (9:30am – 5:30pm), plus a 30-minute lunch.

Topic-by-Topic Delivery Guide

Topic 01: Data Preparation · Slides 10-34

pandas foundations · import/export · selection · filtering · cleaning

- **Analytical workflow:** Frame the decision, acquire data, prepare, analyse, visualise, interpret and review.
- **Series and DataFrame:** Use labelled one- and two-dimensional structures with explicit indexes and data types.
- **Import and export:** Read and write CSV, Excel and JSON while preserving schema, provenance and encoding.
- **Selection:** Use columns, loc and iloc intentionally; distinguish labels from integer positions.
- **Filtering and slicing:** Translate business conditions into reproducible Boolean masks and bounded subsets.
- **Cleaning:** Profile missingness, duplicates, invalid types and ranges before applying documented rules.

Deep-dive coverage:

- **Slide 21 · Analytics, reporting, BI and data science:** These disciplines overlap but answer different needs.
- **Slide 22 · Decision-first analytical framing:** Begin with the decision rather than the available columns.
- **Slide 23 · Observational unit and data grain:** The observational unit explains what one row means; grain adds the time and dimensional detail.
- **Slide 24 · Data quality dimensions:** Quality is fitness for a stated analytical purpose.
- **Slide 25 · Missing data is an analytical signal:** A blank value is not automatically an error and deletion is not automatically neutral.
- **Slide 26 · Duplicate records and business keys:** Duplicates are defined relative to grain and a business key.
- **Slide 27 · Types, categories and encoding:** A storage type is not the same as a measurement meaning.
- **Slide 28 · Reconciliation and an audit trail:** Reconciliation proves that transformation did what was intended.

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- **Slide 29 · Population, sample and selection bias:** A precise calculation can still support the wrong population.

Lab 1: Profile and Clean Retail Transactions · Slides 32-32

Alignment: LO1. Learner output: A clean retail DataFrame, data-quality summary and before/after revenue comparison chart. Acceptance: The script runs without error, clean_data.csv has unique order IDs, critical fields are complete and the quality report reconciles every change.

Lab 2: Select, Filter and Segment Customer Orders · Slides 33-33

Alignment: LO1. Learner output: Reusable customer segments and a chart comparing average order value by segment. Acceptance: The loc and iloc samples are saved, every segment rule is reproducible and the chart uses an honest zero baseline.

Topic 02: Data Transformation · Slides 35-59

concat · merge · sort · groupby · aggregation · pivot tables

- **Concatenate:** Append compatible records by row or combine aligned features by column.
- **Merge:** Join tables on validated keys and inspect cardinality, unmatched records and duplicated rows.
- **Sort:** Order data by one or more fields without confusing row order with analytical importance.
- **GroupBy:** Split data into meaningful segments, apply measures and combine the results.
- **Aggregation:** Use named aggregations so each metric, denominator and output column is explicit.
- **Pivot:** Reshape long and wide data for comparison while retaining the underlying grain.

Deep-dive coverage:

- **Slide 46 · Tidy data and long-form structure:** Tidy data gives pandas operations predictable semantics.
- **Slide 47 · Choose a join from the business question:** A join is a population decision, not merely syntax.
- **Slide 48 · Join cardinality and row multiplication:** Suppose three order rows and two promotion rows share the same product key: a many-to-many join can create six rows.
- **Slide 49 · Unmatched-record audit:** The pandas merge indicator turns join coverage into evidence.
- **Slide 50 · Aggregation grain and denominator:** An aggregation changes grain.
- **Slide 51 · Weighted and unweighted averages:** A mean of group means is valid only when groups should contribute equally.
- **Slide 52 · Rates, proportions and explicit bases:** A rate is interpretable only when its base is visible.
- **Slide 53 · GroupBy, pivot table and crosstab:** Choose the operation from the output grain.
- **Slide 54 · Transformation pipeline controls:** A transformation pipeline should fail visibly when its assumptions are violated.

Lab 3: Integrate Orders, Products and Targets · Slides 57-57

Alignment: LO2. Learner output: An enriched order table, join audit and product-category performance chart. Acceptance: Row counts reconcile, the many-to-one validations pass, unmatched keys are reported and totals before/after integration are explained.

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Lab 4: Summarise Service Operations with GroupBy and Pivot · Slides 58-58

Alignment: LO2. Learner output: A tidy SLA scorecard, priority-by-team pivot and stacked workload chart. Acceptance: The pivot total reconciles, each metric has an explicit denominator and the SLA result is not confused with ticket volume.

Topic 03: Data Visualization · Slides 60-85

pandas plots · Matplotlib · Seaborn · chart choice · visual integrity

- **Question first:** Choose the chart from the comparison, trend, distribution or relationship question.
- **Matplotlib:** Control the figure, axes, labels, annotations and export for reproducible publication.
- **Seaborn:** Create statistical graphics with semantic mappings and consistent themes.
- **Comparison and trend:** Use bars for category magnitude and lines for ordered time.
- **Distribution:** Use histograms and box plots to reveal shape, spread and candidate outliers.
- **Relationship:** Use scatter, regression, pair plots and heatmaps without implying causation.

Deep-dive coverage:

- **Slide 72 · Pre-attentive attributes guide attention:** Viewers detect position, length, colour and size before consciously reading labels.
- **Slide 73 · Chart selection decision tree:** Chart choice follows the analytical question and data types.
- **Slide 74 · Visual encoding accuracy:** Humans compare aligned position and length more accurately than angles, areas or colour intensity.
- **Slide 75 · Zero baselines and honest axes:** A truncated bar axis exaggerates small differences because bar length carries the claim.
- **Slide 76 · Colour accessibility and semantics:** Colour should encode a defined variable or direct attention, not decorate.
- **Slide 77 · Histogram bins change the story:** A histogram is a model of a distribution created by binning continuous values.
- **Slide 78 · Box plot anatomy and limitations:** A box plot compresses a distribution into robust summaries and is useful for comparing groups.
- **Slide 79 · Scatter plots, overplotting and trend:** A scatter plot reveals form, direction, strength, clusters and unusual observations between two numerical variables.
- **Slide 80 · Decision-ready annotation and context:** A decision-ready chart makes its scope verifiable.

Lab 5: Visualise Monthly Retail Performance · Slides 83-83

Alignment: LO3. Learner output: A three-chart management pack: monthly trend, regional comparison and variance waterfall-style bar chart. Acceptance: All three charts render, labels and units are complete, bars use a zero baseline and the rationale connects chart choice to the question.

Lab 6: Explore Marketing Relationships and Correlation · Slides 84-84

Alignment: LO3. Learner output: A relationship-analysis pack with scatter/regression plot, pair plot, heatmap and evidence statement. Acceptance: The graphics render, correlation values are reproducible and the interpretation explicitly separates association from causation.

Topic 04: Data Analysis · Slides 86-113

data types · descriptive statistics · covariance · correlation · time series

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- **Data types:** Distinguish categorical and numerical data, including nominal, ordinal, discrete and continuous.
- **Centre:** Use mean, median and mode according to distribution and decision context.
- **Spread:** Use range, variance, standard deviation and quartiles to quantify variability.
- **Covariance:** Measure whether variables move together, while recognising scale dependence.
- **Correlation:** Standardise linear association from -1 to +1; never treat association as causation.
- **Time series:** Parse datetime, index by time, resample, roll and compare periods without leakage.

Deep-dive coverage:

- **Slide 98 · Population, sample, parameter and statistic:** Descriptive statistics summarise observed data, while inferential reasoning uses a sample to learn about a population.
- **Slide 99 · Measurement scales determine valid analysis:** Measurement scale determines which summaries and operations are meaningful.
- **Slide 100 · Mean, median and mode under an outlier:** For 12, 13, 13, 14 and 48, the mean is 20 while the median and mode are 13.
- **Slide 101 · Quartiles, IQR and the five-number summary:** The five-number summary describes position and spread without assuming a symmetric distribution.
- **Slide 102 · Variance and standard deviation:** Sample variance divides the sum of squared deviations by n−1 to estimate population variability.
- **Slide 103 · Coefficient of variation:** The coefficient of variation expresses standard deviation relative to the mean, making it useful for ratio-scale measures with different magnitudes.
- **Slide 104 · Z-scores and unusual observations:** A z-score standardises a value relative to a chosen reference distribution.
- **Slide 105 · Covariance and correlation:** Covariance depends on measurement scale, making magnitudes difficult to compare.
- **Slide 106 · Pearson or Spearman correlation:** Pearson correlation suits an approximately linear relationship without dominating outliers.
- **Slide 107 · Time-series components:** Time-series observations are ordered and often dependent, so ordinary random comparisons can mislead.
- **Slide 108 · Resampling, rolling windows and leakage:** Resampling changes frequency with an explicit aggregation such as daily maximum or weekly sum.

Lab 7: Analyse Service Quality Distributions · Slides 111-111

Alignment: LO4. Learner output: Descriptive-statistics table, histogram, box plot and evidence-based channel comparison. Acceptance: Statistics reconcile to the source, candidate outliers remain traceable and the interpretation uses median when the distribution is skewed.

Lab 8: Resample and Monitor Energy Demand · Slides 112-112

Alignment: LO4. Learner output: Daily and weekly energy summaries, rolling-demand chart and a peak-demand exception table. Acceptance: The index is complete and sorted, resampled totals are explainable, rolling calculations use no future values and peak days are retained as evidence.

Topic 05: Advanced Data Analytics · Slides 114-138

pipe · apply · custom functions · linear regression · credibility review

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- **Method chains:** Compose readable transformations with assign and pipe instead of hidden intermediate edits.
- **Apply and vectorisation:** Prefer vectorised operations; use apply for justified custom row or column logic.
- **Regression:** Estimate a linear relationship between an outcome and one or more predictors.
- **Residuals and MSE:** Inspect unexplained errors and quantify average squared prediction loss.
- **Interpretation:** Separate coefficient direction, model fit, uncertainty and practical meaning.
- **Credibility review:** Check data quality, assumptions, leakage, limitations and reproducibility before release.

Deep-dive coverage:

- **Slide 126 · Four levels of analytics:** The four levels describe increasing decision ambition, not an automatic maturity ladder.
- **Slide 127 · Correlation, regression and causality:** Correlation and regression can describe patterns and support prediction, but a causal claim requires a credible counterfactual design.
- **Slide 128 · Simple linear regression model:** Simple linear regression fits the line that minimises squared residuals.
- **Slide 129 · Interpreting coefficients responsibly:** A coefficient must be translated with both predictor and outcome units.
- **Slide 130 · Train, test and overfitting:** Overfitting occurs when a model learns sample-specific noise that does not generalise.
- **Slide 131 · Residual diagnostics:** Residuals are the part of the outcome not explained by the fitted model.
- **Slide 132 · MSE, RMSE, MAE and R-squared:** No single metric defines a good model.
- **Slide 133 · Model credibility and communication:** Credibility comes from a chain of evidence rather than a high headline metric.

Lab 9: Build a Reusable pandas Analytics Pipeline · Slides 136-136

Alignment: LO5. Learner output: A reproducible analytics pipeline, validation log and late-delivery risk chart. Acceptance: The pipeline can be re-run from raw input, validation fails loudly on broken schema, assertions pass and every output is derived without manual edits.

Lab 10: Capstone: Public Health Insight and Regression Review · Slides 137-137

Alignment: LO1–LO5. Learner output: A complete capstone pack with clean data, summaries, visual dashboard, linear model, residual review and stakeholder brief. Acceptance: The script reproduces every output, test-set metrics and residuals are reported, causal claims are avoided and limitations are explicit in the stakeholder brief.

Resources Required

- Python 3.11+ or Google Colab with pandas, Matplotlib, Seaborn and scikit-learn.
- Learner Guide and the ten numbered folders under labs/.
- A self-contained lab.py script and synthetic CSV dataset(s) in every lab folder.
- Projector for concept briefings and demonstrations.
- LMS/TMS access for courseware.

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